5.1: Length and Dot Product in $\mathbb{R}^n$

MATH 2360

1. **Length (Norm, or Magnitude) of a Vector:** The length (or norm, or magnitude)of a vector $\vec{v} = (v_1, v_2, \dots, v_n)$ is

$$\|\vec{v}\| = \sqrt{{v_1}^2 + {v_2}^2 + \cdots + {v_n}^2}.$$ ← Ans is #

2. **Unit Vector:** If $\|\vec{v}\| = 1$, then $\vec{v}$ is a unit vector.

3. Each vector in the standard basis for $\mathbb{R}^n$ has length 1 and is a standard unit vector in $\mathbb{R}^n$.

4. **Parallel Vectors:** Two vectors $\vec{u}$ and $\vec{v}$ are parallel when one is a scalar multiple of the other ($\vec{u} = c\vec{v}$). If $c > 0$, they have the same direction; if $c < 0$, they have opposite directions.

5. **Thm: Length of a Scalar Multiple:** Let $\vec{v}$ be a vector in $\mathbb{R}^n$ and let c be a scalar. Then

$$\|c\vec{v}\| = |c| \cdot \|\vec{v}\|.$$

6. **Thm: Unit Direction in Direction of $\vec{v}$:** If $\vec{v}$ is a nonzero vector in $\mathbb{R}^n$, then

$$\vec{U} = \frac{\vec{v}}{\|\vec{v}\|}$$ } Ans is vector

is the unit vector in the direction of $\vec{v}$.

Ex: ← means find norm

1) Find the length of:

a) $\overrightarrow{v_1} = (0, -2, 4, -2)$

$$\|\vec{v}_1\| = \sqrt{0^2 + (-2)^2 + 4^2 + (-2)^2} = \sqrt{0 + 4 + 16 + 4}$$

$$\|\vec{v}_1\| = \sqrt{24} = \sqrt{4 \cdot 6} = 2\sqrt{6}$$

b) $\overrightarrow{v_2} = \left(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}\right)$

$$\|\vec{v}_2\| = \left[\left(\frac{1}{\sqrt{6}}\right)^2 + \left(\frac{2}{\sqrt{6}}\right)^2 + \left(\frac{1}{\sqrt{6}}\right)^2\right]^{1/2} = \sqrt{\frac{1}{6} + \frac{4}{6} + \frac{1}{6}} = \sqrt{\frac{6}{6}}$$

$$= \sqrt{1} = 1$$

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2) Find the unit vector in the direction of $\vec{v} = (-4, 2, 1)$.

↳ $\frac{\vec{v}}{\|\vec{v}\|} = \vec{u}$ ← ans is vector

$\|\vec{v}\| = \sqrt{(-4)^2 + 2^2 + 1^2} = \sqrt{16 + 4 + 1} = \sqrt{21}$

$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \left(\frac{-4}{\sqrt{21}}, \frac{2}{\sqrt{21}}, \frac{1}{\sqrt{21}} \right)$

7. **Distance Between Two Vectors:** The distance between two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ is

substract components ⇒ sq and sum ⇒ √ ← $d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$.

not $= \|\vec{u}\| - \|\vec{v}\| \neq \|\vec{u}\| + \|-\vec{v}\|$

Ex:

3) Find the distance between $\vec{u} = (2, 0, 1, -3)$ and $\vec{v} = (-1, 4, -2, 0)$.

$d(\vec{u}, \vec{v}) = \sqrt{(2+1)^2 + (0-4)^2 + (1+2)^2 + (-3-0)^2}$

$= \sqrt{3^2 + (-4)^2 + 3^2 + (-3)^2} = \sqrt{9 + 16 + 9 + 9}$

$= \sqrt{43}$

8. **Dot Product:** The dot product of $\vec{u} = (u_1, u_2, \ldots, u_n)$ and $\vec{v} = (v_1, v_2, \ldots, v_n)$ is the scalar quantity

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

ans is #

Ex:

4) Find the dot product of $\vec{u} = (2, 0, 1, -3)$ and $\vec{v} = (-1, 4, -2, 0)$.

$\vec{u} \cdot \vec{v} = (2, 0, 1, -3) \cdot (-1, 4, -2, 0)$

$= 2(-1) + (0)(4) + 1(-2) + (-3)(0)$

$= -2 - 2$

$\vec{u} \cdot \vec{v} = -4$

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9. **Thm: Properties of the Dot Product:** If $\vec{u}, \vec{v}, \vec{w}$ are vectors in $\mathbb{R}^n$ and c is a scalar, then:

Know
a) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
b) $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
c) $c(\vec{u} \cdot \vec{v}) = (c\vec{u}) \cdot \vec{v} = \vec{u} \cdot (c\vec{v})$
d) $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$
e) $\vec{v} \cdot \vec{v} \geq 0$, and $\vec{v} \cdot \vec{v} = 0$ iff $\vec{v} = \vec{0}$

Ex:

5) If $\vec{u} = (2, -1), \vec{v} = (3, 4)$, and $\vec{w} = (-5, 2)$, find:

a) $\vec{u} \cdot (2\vec{v}) = (2,-1) \cdot 2(3,4) = (2,-1) \cdot (6,8)$

$= 2(6) + (-1)8 = 12 - 8 = 4$

b) $(\vec{u} \cdot \vec{v})\vec{w}$ # → ans is vector

$\vec{u} \cdot \vec{v} = (2,-1) \cdot (3,4) = 2(3) + (-1)(4) = 6 - 4 = 2$

$(\vec{u} \cdot \vec{v})\vec{w} = 2\vec{w} = 2(-5,2) = (-10,4)$

c) $\|\vec{w}\|^2$ ans # $= (\sqrt{29})^2 = 29$

$\|\vec{w}\| = \|(-5,2)\| = \sqrt{(-5)^2 + 2^2} = \sqrt{25+4} = \sqrt{29}$

d) $\vec{u} \cdot (\vec{v} - 2\vec{w})$

$\vec{v} - 2\vec{w} = (3,4) + -2(-5,2) = (3,4) + (10,-4)$

$= (13,0)$

$\vec{u} \cdot (\vec{v} - 2\vec{w}) = (2,-1) \cdot (13,0) = 2(13) + (-1)(0)$

$= 26$

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6) Consider two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ such that $\vec{u} \cdot \vec{u} = 27, \vec{u} \cdot \vec{v} = -5$, and $\vec{v} \cdot \vec{v} = 51$.
Evaluate $(\vec{u} + 2\vec{v}) \cdot (3\vec{u} + \vec{v})$.

$(\vec{u}+2\vec{v}) \cdot (3\vec{u}+\vec{v}) = \vec{u} \cdot (3\vec{u}+\vec{v}) + 2\vec{v} \cdot (3\vec{u}+\vec{v})$
$= \vec{u} \cdot 3\vec{u} + \vec{u} \cdot \vec{v} + (2\vec{v}) \cdot (3\vec{u}) + 2\vec{v} \cdot \vec{v}$
$= 3(\vec{u} \cdot \vec{u}) + \vec{u} \cdot \vec{v} + 6(\vec{v} \cdot \vec{u}) + 2(\vec{v} \cdot \vec{v})$ Recall $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
$= 3(27) + 7(-5) + 2(51)$
$= 81 - 35 + 102$
$= 148$

10. **Thm: The Cauchy-Schwarz Inequality:** If $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$, then $|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\|\|\vec{v}\|$.

Ex: calculate both sides and make sure ≤

7) Verify the Cauchy – Schwarz Inequality for $\vec{u} = (1, -1, 3)$ and $\vec{v} = (2, 0, -1)$.

$|\vec{u} \cdot \vec{v}| = |(1,-1,3) \cdot (2,0,-1)| = |2+0-3| = |-1| = 1$
$\|\vec{u}\| = \|(1,-1,3)\| = \sqrt{1^2 + (-1)^2 + 3^2} = \sqrt{1+1+9} = \sqrt{11}$
$\|\vec{v}\| = \|(2,0,-1)\| = \sqrt{2^2 + 0^2 + (-1)^2} = \sqrt{4+1} = \sqrt{5}$
$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$
$1 \leq \sqrt{11}\sqrt{5} = \sqrt{55}$ yes

11. The angle θ between two nonzero vectors $\vec{u}, \vec{v}$ in $\mathbb{R}^n$ can be found using

$$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|} \quad (0 \leq \theta \leq \pi).$$

Ex:

8) Find the angle between $\vec{u} = (2, 0, 1, -3)$ and $\vec{v} = (-1, 4, -2, 0)$.

$\vec{u} \cdot \vec{v} = (2,0,1,-3) \cdot (-1,4,-2,0) = 2(-1) + 0 + 1(-2) + 0 = -4$
$\|\vec{u}\| = \sqrt{2^2+0^2+1^2+(-3)^2} = \sqrt{4+1+9} = \sqrt{14} = \sqrt{2 \cdot 7}$
$\|\vec{v}\| = \sqrt{(-1)^2+4^2+(-2)^2} = \sqrt{1+16+4} = \sqrt{21} = \sqrt{3 \cdot 7}$
$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|} = \frac{-4}{7\sqrt{6}} \Rightarrow \theta = \cos^{-1}\left(\frac{-4}{7\sqrt{6}}\right)$

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vectors are ⊥

12. **Orthogonal Vectors:** Two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ are orthogonal when $\vec{u} \cdot \vec{v} = 0$.

$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 = 0$

Ex:

9) Determine all vectors orthogonal to $\vec{u} = (-3, 5)$. Let $\vec{v} = (v_1, v_2)$

$\vec{u} \cdot \vec{v} = (-3, 5) \cdot (v_1, v_2)$

$= -3v_1 + 5v_2 = 0 \Rightarrow 5v_2 = 3v_1 \Rightarrow v_1 = \frac{5}{3} v_2$

$v_2 = t$

$\vec{v} = \left(\frac{5}{3}t, t\right) = t\left(\frac{5}{3}, 1\right) = t(5, 3)$

13. **Thm: The Triangle Inequality:** If $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$, then $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$.

14. **The Pythagorean Theorem:** If $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$, then $\vec{u}$ and $\vec{v}$ are orthogonal iff

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2.$$

The dot product of two $n \times 1$ column vectors $\vec{u}$ and $\vec{v}$ is $\vec{u} \cdot \vec{v} = \vec{u}^T \vec{v}$. ↳ matrix mult

Ex:

10) Find the dot product of $\vec{u} = [2 \ 0]^T$ and $\vec{v} = [3 \ 1]^T$.

$\vec{u} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$\vec{u} \cdot \vec{v} = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = [2(3) + 0(1)] = [6]$

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5.2: Inner Product Spaces

MATH 2360

Definitions:

1. **Inner Product:** Let $\vec{u}, \vec{v}$, and $\vec{w}$ be vectors in a vector space V, and let c be any scalar. An inner product on V is a function that associates a real number $\langle \vec{u}, \vec{v} \rangle$ with each pair of vectors $\vec{u}$ and $\vec{v}$ and satisfies the following axioms:

   a) $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$

   b) $\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$

   c) $c\langle \vec{u}, \vec{v} \rangle = \langle c\vec{u}, \vec{v} \rangle$

   d) $\langle \vec{v}, \vec{v} \rangle \geq 0$, and $\langle \vec{v}, \vec{v} \rangle = 0$ iff $\vec{v} = \vec{0}$

   A vector space V with an inner product is an **inner product space**.

   ↳ general inner product for a vector space

2. **Note:** Since we know

   1) $\vec{u} \bullet \vec{v} = \vec{v} \bullet \vec{u}$,

   2) $\vec{u} \bullet (\vec{v} + \vec{w}) = (\vec{u} \bullet \vec{v}) + (\vec{u} \bullet \vec{w})$,

   3) $c(\vec{u} \bullet \vec{v}) = (c\vec{u}) \bullet \vec{v}$,

   4) $\vec{v} \bullet \vec{v} \geq 0$ from the prior section,

   the dot product is an inner product (known as the **Euclidean inner product**).

Examples

1) Show that $\langle \vec{u}, \vec{v} \rangle = 3u_1v_1 + u_2v_2$ defines an inner product on $\mathbb{R}^2$. Need to show all 4 prop above

a) $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$?

$\langle \vec{u}, \vec{v} \rangle = 3u_1v_1 + u_2v_2 = 3v_1u_1 + v_2u_2 = \langle \vec{v}, \vec{u} \rangle$ ✓

b) $\langle \vec{u}, \vec{v} + \vec{w} \rangle = 3u_1(v_1 + w_1) + u_2(v_2 + w_2)$

$= 3u_1v_1 + 3u_1w_1 + u_2v_2 + u_2w_2$

$= (3u_1v_1 + u_2v_2) + (3u_1w_1 + u_2w_2) = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$ ✓

c) $c\langle \vec{u}, \vec{v} \rangle = c(3u_1v_1 + u_2v_2) = 3(cu_1)v_1 + (cu_2)v_2 = \langle c\vec{u}, \vec{v} \rangle$ ✓

d) $\langle \vec{v}, \vec{v} \rangle = 3v_1v_1 + v_2v_2 = 3(v_1)^2 + (v_2)^2 \geq 0$

b/c anything squared is always positive

$\therefore \langle \vec{u}, \vec{v} \rangle = 3u_1v_1 + u_2v_2$ is an inner product.

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2) Show that $\langle \vec{u}, \vec{v} \rangle = u_1 v_1 - 6u_2 v_2$ does not define an inner product on $\mathbb{R}^2$ Let $\vec{u} = \langle 1, 2 \rangle$

$\langle \vec{u}, \vec{u} \rangle = u_1 u_1 - 6u_2 u_2 = 1^2 - 6(2^2) = 1 - 6(4) = 1 - 24 = -23 < 0$

Breaks prop 4/d

Ex: Let $C[a,b]$ is the set of continuous functions on $[a,b]$ - means all cont fcns on $[a,b]$

3) Show that $\langle f, g \rangle = \int_a^b f(x)g(x)\mathrm{dx}$ defines an inner product on $C[a,b]$ ↑ continuous

a) $\langle f, g \rangle = \int_a^b f(x)\,g(x)\,dx = \int_a^b g(x)\,f(x)\,dx = \langle g, f \rangle$

b) $\langle f, g+h \rangle = \int_a^b f(g+h)\,dx = \int_a^b fg + fh\,dx$

$= \int_a^b fg\,dx + \int_a^b fh\,dx = \langle f, g \rangle + \langle f, h \rangle$

c) $c\langle f, g \rangle = c\int_a^b fg\,dx = \int_a^b cfg\,dx = \langle cf, g \rangle$

d) $\langle f, f \rangle = \int_a^b ff\,dx = \int_a^b f^2\,dx \geq 0$ → a sq. fcn always ≥ 0

If $f = 0$ then $\int f\,dx = 0$

Stop

3. **Thm: Properties of Inner Products:** If $\vec{u}, \vec{v}$, and $\vec{w}$ are vectors in inner product space $\mathcal{V}$ and c is any real number
   1) $\langle \vec{0}, \vec{v} \rangle = \langle \vec{v}, \vec{0} \rangle = 0$
   2) $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$
   3) $\langle \vec{u}, c\vec{v} \rangle = c\langle \vec{u}, \vec{v} \rangle$

4. **Length (Norm), Distance, Angle, and Orthogonality:** Let $\vec{u}$ and $\vec{v}$ be vectors in an inner product space $\mathcal{V}$.
   1) The **length (or norm)** of $\vec{u}$ is $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$.
   2) The **distance** between $\vec{u}$ and $\vec{v}$ is $d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$.

Know

   3) The **angle** between $\vec{u}$ and $\vec{v}$ can be found using $\cos\theta = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\|\|\vec{v}\|}$.
   4) $\vec{u}$ and $\vec{v}$ are **orthogonal** when $\langle \vec{u}, \vec{v} \rangle = 0$.
   5) If $\|\vec{u}\| = 1$, $\vec{u}$ is a **unit vector**, and the unit vector in the direction of $\vec{v}$ is $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$.

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5. **Thm: Cauchy-Schwarz and Triangle Inequalities:** Let $\vec{u}$ and $\vec{v}$ be vectors in an inner product space $\mathcal{V}$. Then

   1) $|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\|\|\vec{v}\|$ (Cauchy – Schwarz Inequality)
   2) $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$ (Triangle Inequality)
   3) $\vec{u}$ and $\vec{v}$ are orthogonal iff $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$

For polynomials: $p = a_0 + a_1 x + \cdots + a_n x^n$ and $q = b_0 + b_1 x + \cdots + b_n x^n$ (only coefficients) $\Rightarrow \langle p, q \rangle = a_0 b_0 + a_1 b_1 + \cdots + a_n b_n$

Examples

1) Let $p(x) = 1 - 2x^2$, $q(x) = 4 - 2x + x^2$, $r(x) = x + 2x^2$ be polynomials in P_2. Find:

   a) $\langle p, q \rangle$ $p(x) = 1 + 0x + -2x^2$ $q(x) = 4 + -2x + x^2$ $r(x) = 0 + 1x + 2x^2$

$\langle p, q \rangle = 1(4) + 0(-2) + (-2)(1) = 4 + 0 + -2 = 2$

   b) $\langle q, r \rangle$

$\langle q, r \rangle = 4(0) + (-2)(1) + (1)(2) = 0 + -2 + 2 = 0$

   c) $\|r\| = \sqrt{\langle r, r \rangle}$ $\|r\| = \sqrt{5}$

$\langle r, r \rangle = 0(0) + 1(1) + 2(2) = 1 + 4 = 5$

   d) $d(p, r) = \|p - r\| = \sqrt{(p_0 - r_0)^2 + \cdots + (p_n - r_n)^2}$

$(p - r)^2 = (1 - 0)^2 + (0 - 1)^2 + (-2 - 2)^2 = 1^2 + (-1)^2 + (-4)^2$
$= 1 + 1 + 16 = 18$

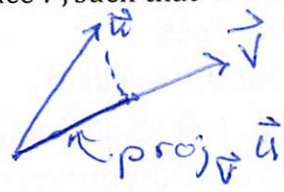
$\|p - r\| = \sqrt{18} = 3\sqrt{2}$

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6. **Orthogonal Projection:** Let $\vec{u}$ and $\vec{v}$ be vectors in an inner product space V, such that $\vec{v} \neq \vec{0}$. Then the orthogonal projection of $\vec{u}$ onto $\vec{v}$ is

$$proj_{\vec{v}}\vec{u} = \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}.$$

ans is vector



Ex:

2) Find the orthogonal projection of $\vec{u} = (5, -3, 1)$ onto $\vec{v} = (1, -1, 0)$ using the Euclidean inner product.

$proj_{\vec{v}}\vec{u} = \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}$

$\langle \vec{u}, \vec{v} \rangle = \langle (5,-3,1), (1,-1,0) \rangle = 5 + 3 + 0 = 8$

$\langle \vec{v}, \vec{v} \rangle = \langle (1,-1,0), (1,-1,0) \rangle = 1 + 1 + 0 = 2$

$proj_{\vec{v}}\vec{u} = \frac{8}{2}(1,-1,0) = 4(1,-1,0) = (4,-4,0)$

7. **Best Approximation (Closest Point) Theorem:** Let $\vec{u}$ and $\vec{v}$ be vectors in an inner space $\mathcal{V}$, such that $\vec{v} \neq \vec{0}$. Then

$$d(\vec{u}, proj_{\vec{v}}\vec{u}) < d(\vec{u}, c\vec{v}) \text{ for any } c \neq \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle}$$

(That is, of all the scalar multiples of $\vec{v}$, the orthogonal projection of $\vec{u}$ onto $\vec{v}$ is the one closest to $\vec{u}$.)

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5.3: Orthonormal Bases: The Gram-Schmidt Process

MATH 2360

Definitions:

1. **Orthogonal / Orthonormal Set:** A set S of an inner product space V is orthogonal when every pair of vectors in S is orthogonal and is orthonormal if each of those vectors is a unit vector.

   If S is a basis, it is an orthogonal (or orthonormal) basis (such as $\{(1,0,0),(0,1,0),(0,0,1)\}$).

To show orthogonal: $\langle \vec{u}, \vec{v} \rangle = 0$ ← only 2 vectors
To show orthonormal: $\|\vec{u}\| = 1$ ← only 1 vector
} both true for orthonormal basis

Examples

1) Show that $S = \left\{\left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}\right), \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)\right\}$ is an orthonormal basis for $\mathbb{R}^3$.

$\vec{u} = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right)$ $\vec{v} = \left(-\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}\right)$ $\vec{w} = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$

$\vec{u} \cdot \vec{v} = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right) \cdot \left(-\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}\right) = -\frac{\sqrt{12}}{12} + 0 + \frac{\sqrt{12}}{12} = 0$

$\vec{u} \cdot \vec{w} = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}\right) \cdot \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right) = \frac{\sqrt{6}}{6} + 0 + \frac{-\sqrt{6}}{6} = 0$

$\vec{v} \cdot \vec{w} = \left(-\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}\right) \cdot \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right) = -\frac{\sqrt{18}}{18} + \frac{\sqrt{18}}{9} + \frac{-\sqrt{18}}{18}$

$= \frac{2\sqrt{18}}{18} + \frac{-2\sqrt{18}}{18} = 0$

$\|\vec{u}\| = \left(\frac{\sqrt{2}}{2}\right)^2 + 0^2 + \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{2}{4} + \frac{2}{4} = \frac{4}{4} = 1$

$\|\vec{v}\| = \left(-\frac{\sqrt{6}}{6}\right)^2 + \left(\frac{\sqrt{6}}{3}\right)^2 + \left(\frac{\sqrt{6}}{6}\right)^2 = \frac{6}{36} + \frac{6}{9} + \frac{6}{36} = \frac{4(6)}{36} + \frac{2(6)}{36}$

$= \frac{6(6)}{36} = 1$

$\|\vec{w}\| = \left(\frac{\sqrt{3}}{3}\right)^2 + \left(\frac{\sqrt{3}}{3}\right)^2 + \left(-\frac{\sqrt{3}}{3}\right)^2 = \frac{3}{9} + \frac{3}{9} + \frac{3}{9} = \frac{9}{9} = 1$

∴ The vectors form an orthonormal space.

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2) The set $S = \{1, \sin(x), \cos(x), \sin(2x), \cos(2x), \ldots, \sin(nx), \cos(nx)\}$ is orthogonal in $C[0, 2\pi]$ with inner product

$$\langle f, g \rangle = \int_0^{2\pi} f(x)g(x)\,dx,$$

$m, n \in \mathbb{Z}^+$, $m \neq n$

Since: *(used in Fourier approximations)*

Trig needed:

$\sin\beta\cos\alpha = \frac{1}{2}[\sin(\beta+\alpha) + \sin(\beta-\alpha)]$

$\sin\beta\sin\alpha = \frac{1}{2}[\cos(\beta-\alpha) - \cos(\beta+\alpha)]$

$\cos\beta\cos\alpha = \frac{1}{2}[\cos(\beta-\alpha) - \cos(\beta+\alpha)]$

$\langle 1, \sin(nx)\rangle = \int_0^{2\pi} \sin(nx)\,dx = \frac{-\cos(nx)}{n}\Big|_0^{2\pi} = -1+1 = 0$

$\langle 1, \cos(nx)\rangle = \int_0^{2\pi} \cos(nx)\,dx = \frac{1}{n}\sin(nx)\Big|_0^{2\pi} = 0 - 0$

$\langle \underset{\beta}{\sin(mx)}, \underset{\alpha}{\cos(nx)}\rangle = \int_0^{2\pi} \sin(mx)\cos(nx)\,dx = \int_0^{2\pi} \frac{1}{2}[\sin(\underset{(m+n)x}{mx+nx}) + \sin(\underset{(m-n)x}{mx-nx})]\,dx$

$= \frac{1}{2}\left[\frac{-\cos((m+n)x)}{(m+n)} + \frac{-\cos(m-n)x}{(m-n)}\right]\Big|_0^{2\pi} = 0$

$\langle \underset{\beta}{\sin(mx)}, \underset{\alpha}{\sin(nx)}\rangle = \int_0^{2\pi} \sin(mx)\sin(nx)\,dx = \int_0^{2\pi} \frac{1}{2}[\cos(mx-nx) - \cos(mx+nx)]\,dx$

$= \frac{1}{2}\left[\frac{\sin(m-n)x}{(m-n)} - \frac{\sin(m+n)x}{m+n}\right]\Big|_0^{2\pi} = 0$

$\langle \underset{\beta}{\cos(mx)}, \cos(nx)\rangle = \int_0^{2\pi} \cos(mx)\cos(nx)\,dx = \frac{1}{2}\int_0^{2\pi} \cos(mx-nx) - \cos(mx+nx)\,dx$

$= \frac{1}{2}\left[\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n}\right]\Big|_0^{2\pi} = 0$

$\therefore$ S is orthogonal in $C[0, 2\pi]$

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2. **Orthogonal Sets are Linearly Independent:** If $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space $\mathcal{V}$, then S is linearly independent.
3. **Corollary:** If $\mathcal{V}$ is an inner product space of dimension n, then any orthogonal set of n nonzero vectors is a basis for $\mathcal{V}$.

Examples

3) Show $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\} = \{(2,3,2,-2), (1,0,0,1), (-1,0,2,1), (-1,2,-1,1)\}$ is a basis for $\mathbb{R}^4$.

$\vec{v}_1$ $\vec{v}_2$ $\vec{v}_3$ $\vec{v}_4$ show $\langle\ \rangle = 0$

$\langle \vec{v}_1, \vec{v}_2 \rangle = \langle (2,3,2,-2), (1,0,0,1) \rangle = 2+0+0+-2 = 0$

$\langle \vec{v}_1, \vec{v}_3 \rangle = \langle (2,3,2,-2), (-1,0,2,1) \rangle = -2+0+4+-2 = 0$

$\langle \vec{v}_1, \vec{v}_4 \rangle = \langle (2,3,2,-2), (-1,2,-1,1) \rangle = -2+6+-2+-2 = 0$

$\langle \vec{v}_2, \vec{v}_3 \rangle = \langle (1,0,0,1), (-1,0,2,1) \rangle = -1+0+0+1 = 0$

$\langle \vec{v}_2, \vec{v}_4 \rangle = \langle (1,0,0,1), (-1,2,-1,1) \rangle = -1+0+0+1 = 0$

$\langle \vec{v}_3, \vec{v}_4 \rangle = \langle (-1,0,2,1), (-1,2,-1,1) \rangle = 1+0+-2+1 = 0$

$\therefore$ Basis b/c all vectors lin combo

4. **Coordinates Relative to an Orthonormal Basis:** If $\mathcal{B} = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is an orthonormal basis for an inner product space $\mathcal{V}$, then the coordinate representation of vector $\vec{w}$ relative to $\mathcal{B}$ is

$$\vec{w} = \langle \vec{w}, \vec{v}_1 \rangle \vec{v}_1 + \langle \vec{w}, \vec{v}_2 \rangle \vec{v}_2 + \ldots + \langle \vec{w}, \vec{v}_n \rangle \vec{v}_n.$$ → This is vector

Examples

4) Find the coordinate matrix of $\vec{w} = (1,2)$ relative to $B = \left\{\left(-\frac{2\sqrt{13}}{13}, \frac{3\sqrt{13}}{13}\right), \left(\frac{3\sqrt{13}}{13}, \frac{2\sqrt{13}}{13}\right)\right\}$.

$\vec{v}_1$ $\vec{v}_2$

$\vec{w} = \langle \vec{w}, \vec{v}_1 \rangle v_1 + \langle w, \vec{v}_2 \rangle v_2$

$= \frac{4\sqrt{13}}{13}\left(-\frac{2\sqrt{13}}{13}, \frac{3\sqrt{13}}{13}\right) + \frac{7\sqrt{13}}{13}\left(\frac{3\sqrt{13}}{13}, \frac{2\sqrt{13}}{13}\right) = \left(\frac{-8\sqrt{13}}{13}, \frac{12\sqrt{13}}{13}\right) + \left(\frac{21\sqrt{13}}{13}, \frac{14\sqrt{13}}{13}\right)$

$\vec{w} = \left(\frac{13\sqrt{13}}{13}, \frac{26\sqrt{13}}{13}\right) = (1,2)$

$[\vec{w}]_B = \begin{bmatrix} \frac{4\sqrt{13}}{13} \\ \frac{7\sqrt{13}}{13} \end{bmatrix}$ Coordinate Basis

since this is what we were give ⇒ coordinate basis

$\langle \vec{w}, \vec{v}_2 \rangle = \langle (1,2), \left(\frac{3\sqrt{13}}{13}, \frac{2\sqrt{13}}{13}\right) \rangle = \frac{3\sqrt{13}}{13} + \frac{4\sqrt{13}}{13} = \frac{7\sqrt{13}}{13}$

$\langle \vec{w}, \vec{v}_1 \rangle = \langle (1,2), \left(\frac{-2\sqrt{13}}{13}, \frac{3\sqrt{13}}{13}\right) \rangle = \frac{-2\sqrt{13}}{13} + \frac{6\sqrt{13}}{13} = \frac{4\sqrt{13}}{13}$

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5. **Gram-Schmidt Orthonormalization Process:**

(1) Let $B = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ be a basis for inner product space V.

(2) Let $B' = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}$, where

coordinate representation of vector

How to find coordinate basis

$$\begin{cases} \vec{w}_1 = \vec{v}_1, \\ \vec{w}_2 = \vec{v}_2 - \dfrac{\langle \vec{v}_2, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 \\ \vec{w}_3 = \vec{v}_3 - \dfrac{\langle \vec{v}_3, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \dfrac{\langle \vec{v}_3, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2 \\ \vec{w}_n = \vec{v}_n - \dfrac{\langle \vec{v}_n, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \dfrac{\langle \vec{v}_n, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2 - \dots - \dfrac{\langle \vec{v}_n, \vec{w}_{n-1} \rangle}{\langle \vec{w}_{n-1}, \vec{w}_{n-1} \rangle} \vec{w}_{n-1} \end{cases}$$

Then B' is an orthogonal basis for V.

(3) Let $u_i = \dfrac{\vec{w}_i}{\|\vec{w}_i\|}$. Then $B'' = \{u_1, u_2, \dots, u_n\}$ is an orthonormal basis for V.

(Also, $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\} = \text{span}\{u_1, u_2, \dots, u_k\}$ for $k = 1, 2, \dots, n$.)

Examples

$\vec{v}_1$ $\vec{v}_2$ $\vec{v}_3$

5) Transform the basis $B = \{(1,0,0), (1,1,1), (1,1,-1)\}$ in $\mathbb{R}^3$ into an orthonormal basis.

$\vec{w}_1 = \vec{v}_1 = (1,0,0)$

$\vec{w}_2 = \vec{v}_2 - \dfrac{\langle v_1, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = (1,1,1) - \dfrac{\langle (1,1,1), (1,0,0) \rangle}{\langle (1,0,0), (1,0,0) \rangle} (1,0,0)$

$= (1,1,1) - \dfrac{1}{1}(1,0,0) = (0,1,1) = \vec{w}_2$

$\vec{w}_3 = \vec{v}_3 - \dfrac{\langle \vec{v}_3, \vec{w}_1 \rangle}{\langle w_1, w_1 \rangle} \vec{w}_1 - \dfrac{\langle \vec{v}_2, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} w_2$

$= (1,1,-1) - \dfrac{\langle (1,1,-1), (1,0,0) \rangle}{\langle (1,0,0), (1,0,0) \rangle} (1,0,0) - \dfrac{\langle (1,1,-1), (0,1,1) \rangle}{\langle (0,1,1), (0,1,1) \rangle} (0,1,1)$

$= (1,1,-1) - \dfrac{1}{1}(1,0,0) - \dfrac{0}{1}(0,1,1) = (0,1,-1)$

$B' = \{(1,0,0), (0,1,1), (0,1,-1)\}$

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6. **Alternative Form of Gram-Schmidt Orthonormalization:** Normalize each $\vec{w}_i$ before using it.

(1) $u_1 = \frac{\vec{w}_1}{\|\vec{w}_1\|} = \frac{\vec{v}_1}{\|\vec{v}_1\|}$

(2) $u_2 = \frac{\vec{w}_2}{\|\vec{w}_2\|}$ where $\vec{w}_2 = \vec{v}_2 - \langle \vec{v}_2, u_1 \rangle u_1$

(3) $u_3 = \frac{\vec{w}_3}{\|\vec{w}_3\|}$ where $\vec{w}_3 = \vec{v}_3 - \langle \vec{v}_3, u_1 \rangle u_1 - \langle \vec{v}_3, u_2 \rangle u_2$

(4) $u_n = \frac{\vec{w}_n}{\|\vec{w}_n\|}$ where $\vec{w}_n = \vec{v}_n - \langle \vec{v}_n, u_1 \rangle u_1 - \ldots - \langle \vec{v}_n, u_{n-1} \rangle u_{n-1}$

Examples

6) Find an orthonormal basis for the solution space of the system ⇒ Solve to find basis ⇒ Find B'

$x_1 + x_2 - x_3 - x_4 = 0$

$2x_1 + x_2 - 3x_3 - 2x_4 = 0$

$\begin{bmatrix} 1 & 1 & -1 & -1 & 0 \\ 2 & 1 & -3 & -2 & 0 \end{bmatrix} \xrightarrow{-2R_1 + R_2 \to R_2} \begin{bmatrix} 1 & 1 & -1 & -1 & 0 \\ 0 & -1 & -1 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 + R_1 \to R_1,\ -R_2 \to R_2}$

$-2(1)+1=-1 \quad -2(-1)+-3=-1 \quad -2(-1)+-2=0$

$\begin{bmatrix} 1 & 0 & -2 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 - 2x_3 - x_4 = 0 \\ x_2 + x_3 = 0 \end{matrix} \Rightarrow \begin{matrix} x_4 = p \\ x_3 = t \\ x_2 = -t \\ x_1 = 2t + p \end{matrix} \Rightarrow \vec{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} p$

$B = \{(2,-1,1,0), (0,0,0,1)\}$ ($\vec{v}_1$, $\vec{v}_2$)

$\|(2,-1,1,0)\| = \sqrt{4+1+1} = \sqrt{6}$

$\vec{u}_1 = \frac{w_1}{\|w_1\|} = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{(2,-1,1,0)}{\sqrt{6}} = \left(\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, 0\right) = \vec{w}_1$

$\langle \vec{v}_2, \vec{u}_1 \rangle = 0$

$\vec{w}_2 = \vec{v}_2 - \langle \vec{v}_2, \vec{u}_1 \rangle \vec{u}_1 = (0,0,0,1) - 0 = (0,0,0,1)$

$\vec{u}_2 = \frac{\vec{w}_2}{\|\vec{w}_2\|} = \frac{0}{1} = 0$

$B' = \left\{\left(\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, 0\right), (0,0,0,1)\right\}$ transition matrix to B

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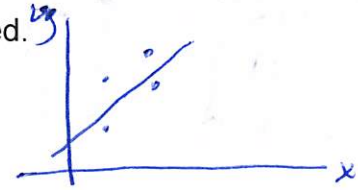
5.4: Mathematical Models and Least Squares Analysis

MATH 2360

When we have an inconsistent system of equations, it is often useful to find the best possible solution, even though we can't find an exact solution.

Definitions and Theorems

1. **Least Squares Problem:** Given an $m \times n$ matrix A and a vector $\vec{b}$ in $\mathbb{R}^m$, the least squares problem is to find $\vec{x}$ in $\mathbb{R}^n$ such that $\|A\vec{x} - \vec{b}\|^2$ is minimized.



2. **Orthogonal Subspaces:** The subspaces S_1 and S_2 are orthogonal when $\vec{v}_1 \cdot \vec{v}_2 = 0$ for all $\vec{v}_1$ in S_1 and all $\vec{v}_2$ in S_2.

Examples

1) Determine whether $S_1 = span\{(-3,0,1)\}$ and $S_2 = span\{(2,1,6), (0,1,0)\}$ are orthogonal.

$\vec{v}_1$ $\vec{u}_1$ $\vec{u}_2$

$\langle \vec{v}_1, \vec{u}_1 \rangle = \vec{v}_1 \cdot \vec{u}_1 = -6 + 0 + 6 = 0$

$\langle \vec{v}_1, \vec{u}_2 \rangle = \vec{v}_2 \cdot \vec{u}_2 = 0 + 0 + 0 = 0$

$\Rightarrow S_1$ orthogonal to S_2

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3. **Orthogonal Complement:** If S is a subspace of $\mathbb{R}^n$, then the orthogonal complement of S is the set $S^{\perp} = \{\vec{u} \in \mathbb{R}^n \mid \vec{v} \cdot \vec{u} = 0 \text{ for all } \vec{v} \in S\}$. (That is, $S^{\perp}=N(A^T)$.)

null space

Ex:

S_1 in next example

2) Find the orthogonal complement $S^{\perp}$ of $S = span\{(0,1,0),(2,0,1)\}$.

$$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{2}e_1 \to e_1} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \Rightarrow \begin{array}{l} x_1 + \frac{1}{2}x_3 = 0 \\ x_2 = 0 \end{array} \qquad \begin{array}{l} x_3 = t \\ x_2 = 0 \\ x_1 = -\frac{1}{2}t \end{array}$$

$$\Rightarrow \vec{x} = \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix} t = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} t \quad \text{(by mult by -2)}$$

$S^{\perp} = \{(1,0,-2)\}$ → S_2 in next example

4. **Direct Sum:** Let S_1 and S_2 be two subspaces of $\mathbb{R}^n$. If each vector $\vec{x}$ in $\mathbb{R}^n$ can be uniquely written as the sum of a vector $\vec{s_1}$ from S_1 and a vector $\vec{s_2}$ from S_2 ($\vec{x} = \vec{s_1} + \vec{s_2}$), then $\mathbb{R}^n$ is a direct sum of S_1 and S_2, and can be written

$$\mathbb{R}^n = S_1 \oplus S_2$$

How to add 2 vector spaces

Ex:

3) From the previous example, write $\mathbb{R}^3$ as a direct sum of S_1 and S_2.

$$S_1 = span\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\right\} \qquad S_2 = span\left\{\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}\right\}$$

$$\mathbb{R}^3 = S_1 \oplus S_2 = span\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}\right\} \oplus span\left\{\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}\right\}$$

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5. **Properties of Orthogonal Subspaces:** If S is a subspace of $\mathbb{R}^n$, then:

Know

(1) $\dim(S) + \dim(S^\perp) = n$

(2) $\mathbb{R}^n = S \oplus S^\perp$

(3) $(S^\perp)^\perp = S$

6. **Projection Onto a Subspace:** If $\{\vec{u_1}, \vec{u_2}, \ldots, \vec{u_k}\}$ is an orthonormal basis for the subspace S of $\mathbb{R}^n$, and $\vec{v}$ in $\mathbb{R}^n$, then

This is a vector → $\text{proj}_S\vec{v} = (\vec{v}\cdot\vec{u}_1)\vec{u}_1 + (\vec{v}\cdot\vec{u}_2)\vec{u}_2 + \ldots + (\vec{v}\cdot\vec{u}_k)\vec{u}_k$.

Ex:

find unit vector

4) Find the projection of $\vec{v} = (2,3,4)$ onto $S = span\{(1,0,1),(0,1,1)\}$.

$\vec{v}_1$ $\quad$ $\vec{u}_1$ $\quad$ $\vec{u}_2$

$$\vec{u}_2 = \frac{(0,1,1)}{\sqrt{1^2+1^2}} = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \qquad \vec{u}_1 = \frac{(1,0,1)}{\sqrt{1^2+1^2}} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

$$\text{proj}_S\vec{v} = (\vec{v}\cdot\vec{u}_1)\vec{u}_1 + (\vec{v}\cdot\vec{u}_2)\vec{u}_2$$

$$(2,3,4)\cdot\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}} + 0 + \frac{4}{\sqrt{2}} = \frac{6}{\sqrt{2}}$$

$$(2,3,4)\cdot\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = 0 + \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{7}{\sqrt{2}}$$

$$\text{proj}_S\vec{v} = \frac{6}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) + \frac{7}{\sqrt{2}}\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{6}{2}, 0, \frac{6}{2}\right) + \left(0, \frac{7}{2}, \frac{7}{2}\right)$$

$$= \left(3, \frac{7}{2}, \frac{13}{2}\right)$$

$$\Rightarrow \text{proj}_S\vec{v} = \begin{bmatrix} 3 \\ 7/2 \\ 13/2 \end{bmatrix}$$

7. **Best Approximation Theorem:** Let S be a subspace of $\mathbb{R}^n$ and let $\vec{v}$ be in $\mathbb{R}^n$. Then for all $\vec{u} \in S$ $(\vec{u} \neq proj_S\vec{v})$, $\|\vec{v} - proj_S\vec{v}\| < \|\vec{v} - \vec{u}\|$. (That is, the orthogonal projection of $\vec{v}$ onto S is the one closest to $\vec{v}$.)

null space

8. **Fundamental Subspaces:** The fundamental subspaces of a matrix A are $N(A), N(A^T)$, (null spaces) $R(A), R(A^T)$ (column spaces).

column space

9. **Thm: Fundamental Subspaces of a Matrix:** If A is an $m \times n$ matrix, then:

(1) R(A) and $\text{N}(\text{A}^\text{T})$ are orthogonal subspaces of $\mathbb{R}^m$.

(2) $\text{R(A)} \oplus \text{N}(\text{A}^\text{T}) = \mathbb{R}^m$

(3) $\text{R}(\text{A}^\text{T})$ and N(A) are orthogonal subspaces of $\mathbb{R}^n$.

(4) $\text{R}(\text{A}^\text{T}) \oplus \text{N(A)} = \mathbb{R}^n$

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10. **Solving the Least Squares Problem:** Now we can solve the least squares problem by finding a vector that minimizes $\|A\vec{x} - \vec{b}\|$. To do this, we solve the normal equations obtained from $A^T A\vec{x} = A^T\vec{b}$.

Ex:

5) Find the least squares solution of $A\vec{x} = \vec{b}$, where A and $\vec{b}$ are given below:

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}$ $\quad \vec{b} = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$ $\quad A^T = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

$A\vec{x} = \vec{b} \Rightarrow A^TA\vec{x} = A^T\vec{b}$

$\begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}\vec{x} = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$

$A^TA = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4+1+1 & 2+2+1 \\ 2+2+1 & 1+4+1 \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix}$

$A^T\vec{b} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 4+0+-3 \\ 2+0+-3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Now, solve $A^TA\vec{x} = A^T\vec{b}$

$\left[\begin{array}{cc|c} 6 & 5 & 1 \\ 5 & 6 & -1 \end{array}\right] \xrightarrow{-R_2+R_1\to R_1} \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 5 & 6 & -1 \end{array}\right] \xrightarrow{-5R_1+R_2\to R_2} \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 11 & -11 \end{array}\right] \xrightarrow{\frac{1}{11}R_2\to R_2}$

$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 1 & -1 \end{array}\right] \xrightarrow{R_2+R_1\to R_1} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array}\right]$

$\Rightarrow \vec{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

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6) Find the projection of $\vec{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ onto the column space of $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$.

$A^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$A^TA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0+1 & 0+0+1 \\ 0+0+1 & 0+1+1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$A^T\vec{b} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2+0+4 \\ 0+3+4 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$

$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\vec{x} = \begin{bmatrix} 6 \\ 7 \end{bmatrix} \Rightarrow \left[\begin{array}{cc|c} 2 & 1 & 6 \\ 1 & 2 & 7 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 2 & 1 & 6 \end{array}\right]$

$\xrightarrow{-2R_1+R_2 \to R_2} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -3 & -8 \end{array}\right] \xrightarrow{-\frac{1}{3}R_2 \to R_2} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 8/3 \end{array}\right] \xrightarrow{-2R_2+R_1 \to R_1}$

$-2(2)+1 = -3 \qquad -2(7)+6 = -14+6 = -8$

$-2\left(\frac{8}{3}\right)+\frac{21}{3} = \frac{-16}{3}+\frac{21}{3} = 5/3$

$\left[\begin{array}{cc|c} 1 & 0 & 5/3 \\ 0 & 1 & 8/3 \end{array}\right] \Rightarrow \vec{x} = \begin{bmatrix} 5/3 \\ 8/3 \end{bmatrix}$

$\text{proj}_S\vec{b} = A\vec{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 5/3 \\ 8/3 \end{bmatrix} = \begin{bmatrix} 5/3+0 \\ 0+8/3 \\ 5/3+8/3 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 8/3 \\ 13/3 \end{bmatrix}$

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